

1. (4 points) Filters at a water treatment plant next to a lake become less effective over time, letting more pollution pass through each day. The rate at which pollution passes through the filters and into the lake is given in the following table.

Time (days)	0	6	12	18	24	30
Rate (kg/day)	7	8	10	13	17	35

By using either a left-hand sum or a right-hand sum, find an **underestimate** for the total quantity of pollution that entered the lake during this 30-day period. Show your calculation, including units.

The total quantity of pollution entering the lake was at least

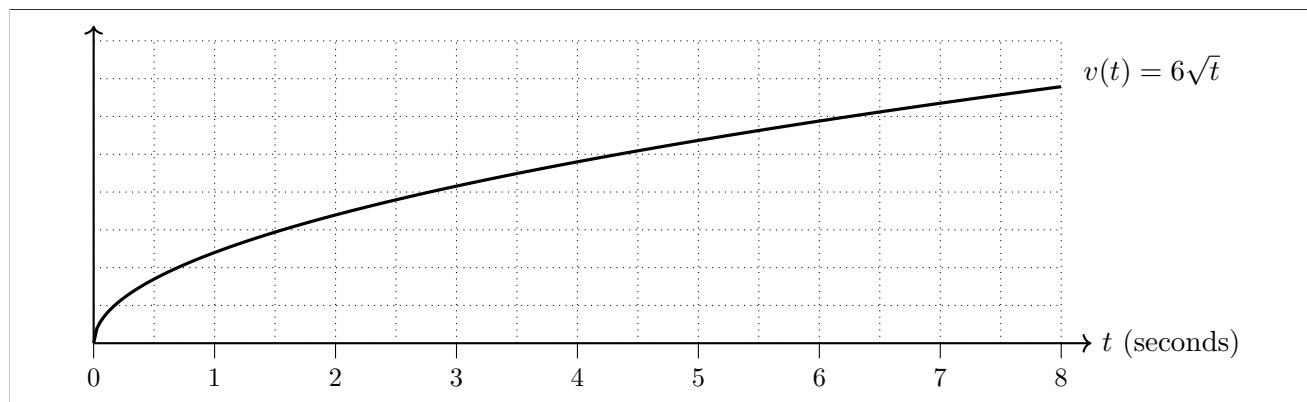
(number)	(units)

2. (5 points) Charles the cougar runs in one direction along a straight path, with velocity $v(t) = 6\sqrt{t}$ in metres per second, where t is the time in seconds since he started running.

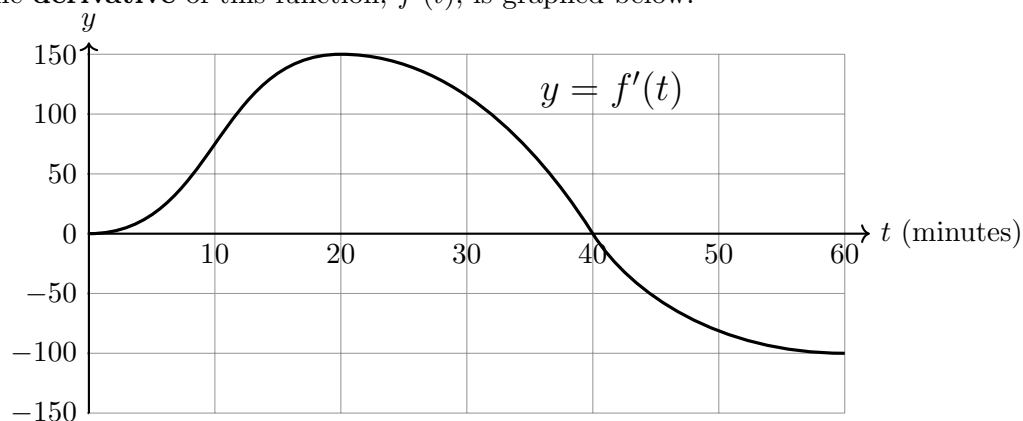
Which of the following is a **right-hand sum** that estimates the distance that Charles travelled from time $t = 2$ seconds to $t = 4$ seconds? Select one correct answer and sketch the sum below.

☐ $\sum_{i=0}^3 6\sqrt{\frac{i}{2}} \cdot \frac{1}{2}$
☐ $\sum_{i=1}^4 6\sqrt{\frac{i}{2}} \cdot \frac{1}{2}$
☐ $\sum_{i=2}^4 6\sqrt{\frac{i}{2}} \cdot \frac{1}{2}$
☐ $\sum_{i=4}^8 6\sqrt{\frac{i}{2}} \cdot \frac{1}{2}$
☐ $\sum_{i=5}^8 6\sqrt{\frac{i}{2}} \cdot \frac{1}{2}$

Illustrate the sum on the graph below.



3. (10 points) Phileas Fogg takes a 1-hour flight in a hot air balloon. Let $f(t)$ represent the altitude (in metres) of the hot air balloon, relative to sea level, at time t (in minutes) since taking off from the ground. The **derivative** of this function, $f'(t)$, is graphed below.



- (a) At about what time was the balloon rising most quickly?

☐ 0 min ☐ 10 min ☐ 20 min ☐ 30 min ☐ 40 min ☐ 50 min ☐ 60 min

- (b) At about what time was the balloon located highest above sea level?

☐ 0 min ☐ 10 min ☐ 20 min ☐ 30 min ☐ 40 min ☐ 50 min ☐ 60 min

- (c) If Phileas started his flight from a beach next to a sea, which of the following might describe his destination at the end of the hour, based on the graph? Please select **all** possible correct answers.

☐ on top of a hill ☐ back on the beach ☐ in the sea

- (d) Another MAT133 student has approached you for help with part (c). Explain your answer to part (c) carefully, to help them understand your reasoning.

4. (10 points) You are providing consultation to government representatives on oil production from a well in Oil Springs, Ontario. Let $r(t)$ represent the rate (in barrels per day) at which oil is pumped from the well at time t in days since the beginning of January 2024. Let $R(t)$ be an antiderivative of $r(t)$.

You've gathered the following data:

$$\begin{array}{lll} r(0) = 170 & r(7) = 165 & r(26) = 160 \\ R(0) = 5000 & R(7) = 6300 & R(26) = 9600 \end{array}$$

Evaluate each quantity below, show your reasoning, and then interpret your answers in non-technical language for the government representatives, who have not taken a calculus course before.

(a) $\int_0^7 r(t) dt =$

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(number)
(units)

Find the value. Show your reasoning. Then write your answer in the box above.

Interpret the value in non-technical language for the representatives.

(b) $r(10) + \int_{10}^{26} r'(t) dt =$

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(number)
(units)

Find the value. Show your reasoning. Then write your answer in the box above.

Interpret the value in non-technical language for the representatives.

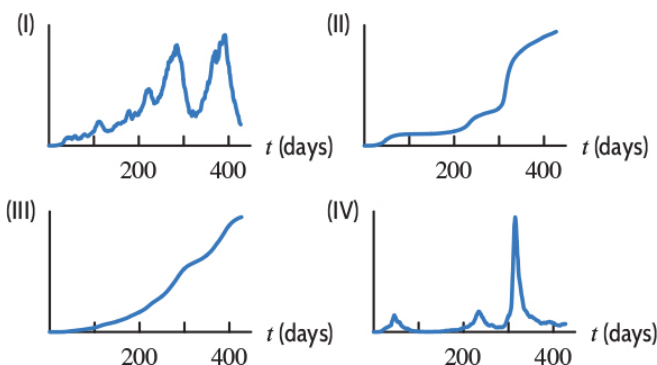
5. (5 points) As we analyze Covid-19 data in a country, we define the following functions:

- Let $N(t)$ be the number of new Covid-19 cases confirmed during day t .
- Let $P(t)$ be the total number of Covid-19 cases confirmed during or before day t .

(a) Which of the following statements is approximately true?

☐ $P(t)$ is an antiderivative of $N(t)$. ☐ $N(t)$ is an antiderivative of $P(t)$.

(b) Real Covid-19 data sets from two different countries are represented in the graphs below. For each country, there is one graph of $P(t)$ and one graph of $N(t)$.



Please fill in only one bubble in each box below to make the statements correct. You may speak about the two countries in either order.

For one country,	☐ (I) ☐ (II) ☐ (III) ☐ (IV)	is the graph of $N(t)$ and	☐ (I) ☐ (II) ☐ (III) ☐ (IV)	is the graph of $P(t)$.
For the other,	☐ (I) ☐ (II) ☐ (III) ☐ (IV)	is the graph of $N(t)$ and	☐ (I) ☐ (II) ☐ (III) ☐ (IV)	is the graph of $P(t)$.

6. (4 points) Gidgette runs a small business that produces luxury vegan facial serum. She has modelled her marginal cost as a function of the quantity q of serum (in litres) produced as

$$MC(q) = 3q^2 - 12q + 24.$$

If her cost for producing 10 litres of serum is \$700, what is her fixed cost for producing serum?

Gidgette's fixed cost for producing luxury vegan facial serum is \$.

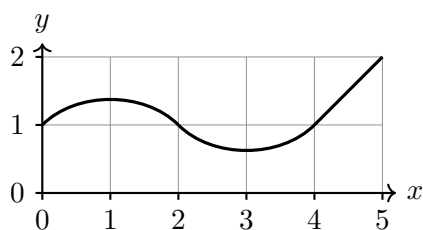
7. (4 points) Evaluate the integral $\int_0^3 x e^{\frac{1}{3}x} dx$. Show your reasoning. Round to the nearest 0.01 if needed.

Answer: $\int_0^3 x e^{\frac{1}{3}x} dx =$

8. (3 points) Suppose that the function f satisfies $\int_0^2 f(t) dt = 3$. Evaluate the integral $\int_0^4 f\left(\frac{t}{2}\right) dt$.

Answer: $\int_0^4 f\left(\frac{t}{2}\right) dt =$

9. (2 points) Consider the function graphed below.



- (a) What is the average value of this function on the interval from 0 to 4?

Answer: The average value is

- (b) What is the average value of this function on the interval from 0 to 5?

☐ 1 ☐ 1.1 ☐ 1.5 ☐ 5 ☐ 5.1 ☐ 5.5 ☐ Other:

10. (9 points) Bob's Big Boy restaurant is considering buying a mochi donut machine that costs \$10,000. The machine will generate a continuous income stream, at a constant rate of c dollars per year. The machine will need to pay for itself in five years, i.e. the future value of its income stream in five years must be greater than or equal to the future value of today's \$10,000 payment in five years. Otherwise, the shareholders will consider it a bad investment and fire the CEO. The restaurant uses a savings account with an interest rate of 4% compounded continuously.

- (a) If the machine is purchased today and generates income continuously at a rate of $c = 2000$ dollars per year for 5 years, what is the **present value** of the income stream from the machine?

The present value of the machine's income stream is \$

(Please round to the nearest \$0.01.)

- (b) If the machine is purchased today and generates income continuously at a rate of $c = 2000$ dollars per year for 5 years, will the CEO be fired? Explain your answer below.

☐ Yes, the CEO will be fired. ☐ No, the CEO will not be fired.

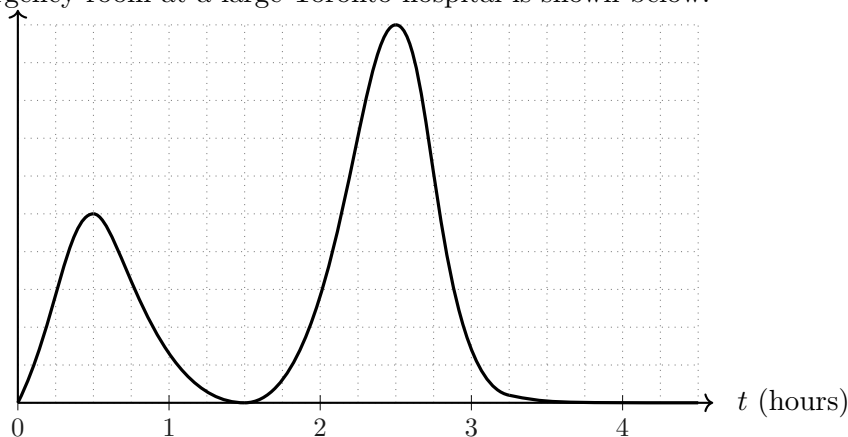
Explanation:

- (c) If the machine is purchased today, at least what rate c must it earn income in order to ensure that the CEO keeps their job? Show your reasoning.

The machine must earn income at a rate of at least $c =$ dollars per year.

(Please round to the nearest \$0.01.)

11. (10 points) The density function, $p(t)$, for the time t (in hours) a patient waits before being seen by a doctor in the emergency room at a large Toronto hospital is shown below.



(a) What are the units on the vertical axis?

(b) In plain language, what does $p(0.75) = 0.3$ mean, approximately? Choose the best answer.

- ☐ 75% of patients wait exactly 0.3 hours.
- ☐ 30% of patients wait exactly 0.75 hours.
- ☐ 75% of patients wait at most 0.3 hours.
- ☐ 30% of patients wait at most 0.75 hours.
- ☐ 0.75% of patients wait between 0.30 and 0.31 hours.
- ☐ 0.3% of patients wait between 0.75 and 0.76 hours.
- ☐ None of the above.

(c) Determine whether each statement about the emergency room is true or false.

Most patients wait more than an hour in the emergency room.

☐ True ☐ False

Essentially no patient sees the doctor within the first 15 minutes of arrival to the ER.

☐ True ☐ False

Essentially all patients have to wait either exactly 0.5 hours or exactly 2.5 hours.

☐ True ☐ False

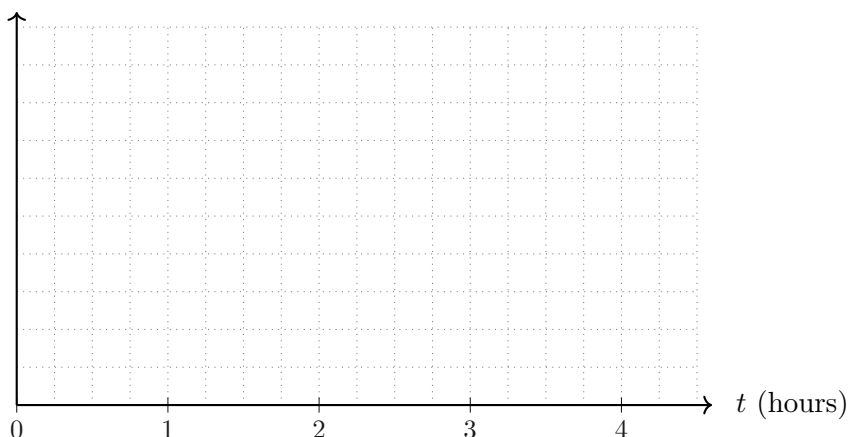
Essentially no patients have to wait more than 4 hours.

☐ True ☐ False

The median wait time is about 1.5 hours.

☐ True ☐ False

(d) Draw a graph of the **cumulative distribution function** (cdf) for the emergency room wait times. Be sure to label the vertical axis and label any key features of your graph.



12. (3 points) Speedy Steve's is a car rental agency. The cost, C , (in dollars) of renting a car at Speedy Steve's is determined by the time t (in days) of the rental and the distance x (in kilometres) that the car is driven.

$x \backslash t$	2	4	6	8	10
50	45	85	125	165	205
100	50	90	130	170	210
150	55	95	135	175	215
200	60	100	140	180	220

- (a) Mr. Bean plans to rent a car for 4 days and drive it 150 km. How much will it cost Mr. Bean to rent the car from Speedy Steve's?

It will cost Mr. Bean dollars to rent the car from Speedy Steve's.

- (b) Based on the table, which of the following could be the formula for the cost C of renting the car at Speedy Steve's as a function of t and x ? Choose one correct answer.

☐ $C = 20t + 0.1x$
☐ $C = 20t - 0.1x$
☐ $C = -20t + 0.1x$
☐ $C = -20t - 0.1x$
☐ $C = 0.1t + 20x$
☐ $C = -0.1t + 20x$
☐ $C = 0.1t - 20x$
☐ $C = -0.1t - 20x$

13. (1 point) Please take a moment to reflect. There are no right or wrong answers to this question, however, you will receive a point for answering fully.

(a) Which question did you feel the most confident about solving?

(b) Which question did you find the most challenging?

- (c) Before you leave the test room, please check all of your work. Find at least one solution that you can make an improvement on. Briefly describe the improvement you made below.

I made an improvement in my solution to Question . My improvement was...

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